

Tusher Bhomik

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EXPERIENCE

- **ACI Limited** August 2026 – Present
Machine Learning Engineer Dhaka, Bangladesh

EDUCATION

- **Bangladesh University of Engineering and Technology** January 2022 – June 2026, CGPA: 3.3164/4.00
Bachelor of Science in Computer Science & Engineering Dhaka, Bangladesh
- **Notre Dame College** 2018 – 2020, GPA: 5.00/5.00
Higher Secondary Certificate (HSC), Science Dhaka, Bangladesh

RESEARCH

- **Improving the Function Calling Capability of Large Language Models** 2026 – Present
Supervisor: Md. Rizwan Parvez, Qatar Computing Research Institute Doha, Qatar
 - Built **synthetic training data** for **multi-hop tool use** after inference-time methods (output post-processing, **RAG** tool retrieval) fell short on multi-step calls.
 - Fine-tuned **Qwen 3** and **Gemma 4** with **SFT and RL (GRPO and DPO-variant objectives, step-wise rewards)**; evaluated on **BFCL** against proprietary baselines.
 - Did not close the gap; traced the cause to synthetic data that **under-covered domains**, limiting generalization to unseen tools.
- **Agentic Security and Failure Analysis of LLM Agent Systems** 2026 – Present
Supervisor: Md. Rizwan Parvez, Qatar Computing Research Institute Doha, Qatar
 - Studying **secure code execution by LLM agents** and *why multi-agent systems fail*, building a **taxonomy** of failure modes.
 - Building a **harder agentic-security benchmark** on open models (Gemma 4, Qwen 3), toward training a security-aware agent model on it.
- **Age-Aware Neurological Gait Pattern Discovery from Biomechanical Signals: Multi-Phase Stroke Severity Analysis** 2025 – 2026
Undergraduate Thesis (defended) | Supervisor: Mahmuda Naznin, Dept. of CSE, BUET Dhaka, Bangladesh
 - An **age-normative framework** for uncovering neurological relations from biomechanical gait features using **machine learning with explainable AI**.

PROJECTS

- **HealthSync** 2025
Spring Boot, React, PostgreSQL, Firebase, Docker, Azure, CI/CD [GitHub](#)
 - Developed a comprehensive healthcare platform connecting patients and doctors with appointment scheduling, digital prescription management, and medical history tracking.
 - **Containerized** the services with **Docker** and deployed to **Microsoft Azure** through a **CI/CD pipeline**.
- **Arxiv Research Paper RAG System** 2026
Python, SQL, SQLite, FastAPI, ChromaDB, Data Analysis [GitHub](#)
 - Built an end-to-end **data pipeline** for ingesting, cleaning, and analyzing arXiv research paper data using Python and SQL.
 - Developed a **RAG-based API system** using FastAPI and ChromaDB for semantic retrieval and question-answering.
- **Enhanced xv6-riscv Kernel** 2024
C, Assembly, Makefile [GitHub](#)
 - Implemented an **MLFQ scheduler** with aging, added **kernel-level threading**, and built custom system calls with a Bash autograder.
- **Mini C Compiler** 2024
CSE-310 Compiler Design, C, 8086 Assembly [GitHub](#)
 - Built a **mini C compiler** from scratch (symbol table, lexical and semantic analysis, intermediate code generation), emitting **8086 assembly**.
- **Computer Graphics: Rasterizer & Ray Tracer** 2025
C++, OpenGL [GitHub](#)
 - Built a software **rasterization pipeline** and a **ray tracer** from scratch, alongside OpenGL scene demos.

ACHIEVEMENTS

- **5th Place** — **Tech Career Hunt 2.0**, organized by Shohoj Coding.
- **Selected for AI/ML Internship** — **Save the Children**, through their Hackathon (AI/ML Track).

May 2026

SKILLS

- **Research Interests:** LLMs, LLM Agents, RL Post-Training, Synthetic Data Generation, Evaluation & Benchmarking, Trustworthy AI, AI Security
- **Programming:** Python, C++, C, Java, JavaScript
- **ML/AI:** PyTorch, Hugging Face Transformers, vLLM, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV
- **Backend:** FastAPI, Spring Boot, Express.js
- **Frontend:** React, Next.js, Material-UI
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools:** Git, Docker, Linux

ADDITIONAL INFORMATION

Languages: English, Bengali. **Interests:** Photography, Traveling, Badminton, Cycling.

REFERENCES

- **Prof. Dr. Mahmuda Naznin** — Thesis Supervisor
Department of CSE, Bangladesh University of Engineering and Technology
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